Highlights

WS4: a compiled-state software architecture with exact replay for landscape harvest scheduling

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- Compiled-state architecture separates model, exact state, and search.
- Public TSA24 case: five seeds with exact CPU replay.
- Microbenchmarks are bounded, not architecture claims.

WS4: a compiled-state software architecture with exact replay for landscape harvest scheduling

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ARTICLE INFO

Keywords:
landscape harvest scheduling
compiled model
GPU population search
exact replay
reproducibility

ABSTRACT

WS4 is an open-source Python system that compiles landscape harvest-scheduling models into a stable sparse form and searches them with exact accounting. It separates four concerns usually rebuilt together: a versioned compiled model, compact candidate and sparse effect indexes, exact resident accounting, and batched GPU search with exact CPU replay. Every reported schedule is replayed through the exact CPU evaluator that is the evidence authority. On a public frozen TSA24 landscape with 208,255 operable blocks, a 100,000-generation five-seed campaign reaches 512.574–518.621 (mean 515.682), and all five schedules replay exactly. The level penalty is a normalized mean-level term, not a per-period floor: the realized level ranges from 51.21 to 52.49M m^3 per 10-year period against a deliberately unattainable 70M aspiration. A 5,000-block diagnostic reduces move-evaluation time by 2.00×10^3 with objective agreement to 3.7×10^{-9}, and a 64-genome factorized batch runs 248 times faster on CUDA than on a single-device CPU path. The contribution is the software architecture and its replay contract; we make no independent harvest-scheduling comparison or state-of-the-art claim.

1. Motivation and significance

Landscape harvest scheduling assigns treatments to blocks over time under growth, operability, and flow objectives. At timber-supply-area scale the state is large and the objective is a weighted compromise, so exact methods are limited and heuristic search is standard [1, 2]. The recurring software problem sits beneath the optimisation problem. A compiled forest model contains blocks, fragments, strata, tracks, treatments, transitions, growth curves, products, and accounting rules. A search move changes one small part of that model, but implementations often rebuild a much larger state to evaluate it. The model, the accounting state, and the move are therefore rebuilt together, which makes experiments slow, replay fragile, and the effect of a search method hard to separate from an implementation detail.

Harvest-scheduling studies have compared search methods for decades: simulated annealing with spatial constraints [3], tabu search against a genetic algorithm [4], and eight heuristic techniques [5]. Reviews document the range of spatial formulations [6, 7], and exact approaches remain restricted [8]. These studies treat the search method as an experimental factor, but each is tied to its own model and accounting implementation. Reproducibility guidance makes the same point: a result depends on the compiled model, candidate order, weights, hardware, and replay procedure, not only on a seed and an objective value [9, 10, 11, 12].

WS4 (Wood Supply Simulation System) addresses that layer. It compiles a landscape model into versioned columnar tables, builds a compact candidate index and a sparse effect table, maintains an exact resident accounting state, runs a batched GPU population search, and replays every reported schedule through an exact CPU evaluator. The software is developed by the UBC FRESH Lab. This paper is a software note: it describes the design and the exact-state contract, illustrates them on one public instance, and states what the evidence does and does not establish. It makes no environmental-modelling claim, no independent harvest-scheduling comparison, and no state-of-the-art performance claim.

The intended users are groups with a compiled spatial model who want to vary search operators without rebuilding accounting, reviewers who need to replay an incumbent, and instructors who need an inspectable example of exact incremental evaluation.

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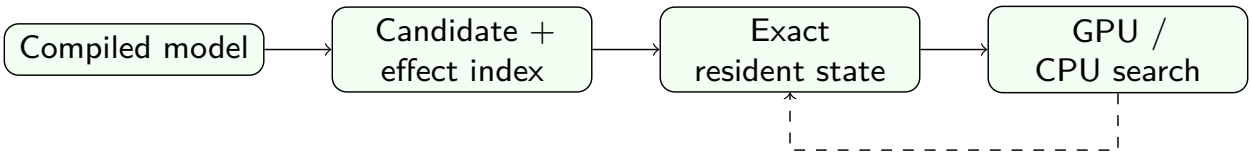


Figure 1: Compiled-state architecture. The resident state is authoritative; search reads and writes it through exact apply and undo operations.

2. Software description

2.1. Architecture overview

The software has four layers (Figure 1). A compiled model is the stable input. A candidate index and a sparse effect table make it cheap to query. An exact resident state holds the realized accounting totals and is the objective authority. A batched GPU population search explores assignments, and an exact CPU replay path verifies the reported result. The design also supports a CPU-refinement handoff, in which an elite schedule is loaded into the resident state, improved with local moves, and written back as a GPU seed; that handoff is designed but is not part of the measured campaign.

The layers exchange explicit objects rather than implicit global state. The search sees only the compiled tables, the candidate list, the sparse rows, and the resident state, so it cannot read raw inventory or mutate accounting outside the apply/undo interface. A run builds or loads the index and table, initializes the state, runs for a declared budget, writes a record, and replays it through the exact evaluator. Each stage writes an artifact with an identity, so the next stage can reject a mismatch.

2.2. Compiled model and candidate index

The compiled dataset is the boundary between ingestion and execution. It is a versioned directory of columnar tables plus a manifest. The tables describe fragments, blocks, tracks, strata, treatments, curves, features, products, accounts, groups, and compile diagnostics. The manifest records the schema version, source identities and hashes, compile options, and table list. Compilation happens at the fragment level; a block is a runtime aggregate of its fragments. The search never reads the raw inventory or recompiles the recipe. Compile diagnostics, such as blocks with no operable chain and unreachable transitions, are stored as data so an experiment can state how many blocks were excluded and why. Compile options are part of the identity because eligibility and operability windows change the candidate set; two datasets with the same source but different options are different models.

For each block, the candidate index stores the feasible treatment chains in a stable order. A chain is an ordered sequence of treatment events, each a treatment and a year; the empty chain is included and is a valid decision. The index is built by walking track transitions from the block's initial state and applying the compiled windows. The order is part of the contract: the effect table and the search reference candidate positions, so changing the order changes the meaning of a cached effect. The index is smaller than the full cross-product but richer than a binary harvest decision.

2.3. Sparse effect table

A block/chain pair affects only a small number of `(year, account)` cells. The effect table stores, for each block and candidate chain, a sparse row of those cells and their values relative to the untreated block, plus the empty-schedule base totals and block row starts. A move is therefore priced by the cells that the old and new chains touch, not by iterating over the whole schedule. Rows are relative, so apply and undo are an addition and a subtraction rather than a recomputation, and the base is stored once rather than duplicated for every candidate.

The table has three costs. The build cost is proportional to the number of candidate chains times the cells per chain and is paid once per dataset and table format. The storage cost is proportional to the number of non-zero cells, not to the dense block-by-period-by-account product. The query cost for one move is proportional to the old and new rows. The table identity includes the dataset fingerprint, candidate order, target-cell geometry, horizon, period width, and table format version. It is a cache key: a dataset change forces a rebuild, while a change to weights, priorities, or targets is a pricing change and reuses the table.

The identity check is also a failure detector, because a stale table would attach rows to the wrong model or chains. If the table does not fit in device memory, a population search cannot run; the TSA24 table uses a lean format that omits the full row index.

2.4. Exact resident state and objective

The resident state stores realized totals per `(period, account)` plus the current chain for each block. Applying a move subtracts the old row and adds the new row; undo applies the inverse. The invariant is that the state always equals the empty-schedule base plus the rows of the current assignment. The update does not depend on move history, so a block that reached a chain after many toggles contributes exactly what a block that reached it in one step contributes. Compound moves are decomposed into elementary changes before they reach the state, and the state never stores a search-specific diff.

The objective is weighted goal programming. Each target contributes a one-sided penalty for falling below a minimum or exceeding a maximum:

$$s_{t,p} = \max(m_t - v_{t,p}, 0), \tag{1}$$

$$e_{t,p} = \max(v_{t,p} - M_t, 0), \tag{2}$$

$$\tau_{t,p} = w_t^- \, \phi(s_{t,p}) + w_t^+ \, \phi(e_{t,p}), \tag{3}$$

where ϕ is linear or quadratic, $v_{t,p}$ is the realized value of target t in period p, and m_t and M_t are the hinge values. The target set is not a set of hard constraints: "minimum" and "maximum" name the hinge of the penalty, not a feasible interval.

The TSA24 target has two roles. One even-flow target spans periods 1–20, compares each period with the reference period, and charges a squared deviation outside a ±1% band with weight 10,000. Second, the level target is a normalized linear penalty on the mean realized level over the twenty periods:

$$\ell = 20 \times 100 \times \frac{70{,}000{,}000 - \bar{v}}{70{,}000{,}000}, \tag{4}$$

where $\bar{v}$ is the mean realized level in m^3 per 10-year period and 70,000,000 m^3 is the aspiration. The level term is dimensionless and is scaled by the number of periods; it is not a per-period hard floor. The aspiration is deliberately above the level the estate is expected to achieve, so it creates a common pull-up gradient. The even-flow target selects the flattest profile at the level where the two roles reach their best weighted compromise. The realized level is an outcome, not a declared feasible floor.

For a move on block b from chain c_{old} to chain c_{new}, let p be the sparse pending change to the realized account totals and let $\mathcal{T}$ be the targets it affects. The objective delta is

$$\Delta Z_b(c_{\text{old}} \to c_{\text{new}}) = \sum_{t \in \mathcal{T}} \left[\pi_t(T + p) - \pi_t(T) \right], \tag{5}$$

where π_t is the target's penalty function and T is the current state. For a linear level target, π_t is a per-cell penalty. For the even-flow target, π_t is a penalty over the period vector, so the delta re-evaluates the affected flow target rather than summing independent cell penalties. The cost is proportional to the pending cells plus the flow target's period vector, not to the number of blocks in the schedule. The implementation uses this identity in the local-search paths and retains a whole-schedule evaluator as a reference.

A schedule is feasible when every operable block has one candidate chain and the chain's events obey the compiled windows; a feasible schedule can still miss every target, because the objective is a weighted penalty, not a feasibility measure. The search uses four move families: toggle a chain, shift an event, swap chains between blocks, and destroy-and-repair a small set. The first three are local and reversible; the fourth is decomposed into local changes. The replay contract applies to the final assignment, not the move decomposition.

2.5. GPU population search and exact replay

The GPU population evaluates a batch of candidate-index genomes against the sparse table. The batch is a matrix of block/chain choices; the kernel accumulates the corresponding rows into a per-genome objective.

The device path produces a ranking and does not update the resident state. The CPU path reconstructs the state from the stored assignment and computes the objective with the same accounting rules. The two paths can differ in floating-point order, so the record stores the CPU replay objective, not the device objective. A small device/CPU difference is acceptable as a search signal; it is not acceptable as a reported result.

Every reported schedule is stored as a complete block/treatment/offset assignment. The record carries the instance and its availability, the objective and feasibility definition, the algorithm, version, and configuration, the seed, the hardware and software, the wall and search times, the schedule path and SHA-256, the exact-replay flag, and the per-target residuals. A validator checks the schema, the schedule path, and the hash. The replay gate is a separate program: it loads the compiled model, reads the stored schedule, evaluates the objective through the compiled model, and checks the record fields without calling the optimizer. It verifies the incumbent, not the search trajectory, and does not prove that the search explored the neighbourhood it claims.

The software therefore supports two levels of reproducibility. The search is reproducible from its configuration and seed for a fixed implementation, and the reported result is replayable from the stored schedule even if the search is not rerun. The record format also permits a table to be regenerated from the record files rather than typed by hand. Four failure modes are exposed: a model change without a table rebuild, an irreversible move, a reported device objective, and a partial or infeasible schedule. None is silent in a record that passes the gate. The gate does not catch a bad search design: a poor schedule can still produce an exact, replayable record. The harness guarantees the evidence; the experiment argues about the search.

2.6. Implementation and use

The implementation is in Python 3.12 and uses NumPy [13], SciPy [14], Numba for optional acceleration [15], and PyTorch for the GPU path [16]. The reported campaign ran on an NVIDIA RTX PRO 6000 Blackwell Max-Q Workstation Edition (97,887 MiB) with an Intel Xeon Gold 6254 CPU at 3.10 GHz, Python 3.12.3, and WS4 release candidate `ws4-v0.2.0-p14rc1` [17]. The command-line surface is small: compile a dataset, build an effect table, run a search with a configuration file, and replay a record. The test suite covers the compiled-table schema, candidate index, effect rows, apply/undo invariants, evaluator, and record validator. The large GPU effect table and caches are not part of the small public dataset artifact; they are rebuilt from the published recipe.

3. Validation and illustration

3.1. Instance and target

The case study uses the public frozen artifact `ws4-tsa24-evenflow-v1`, a TSA24/Prince George landscape compiled for WS4 [18]. The compiled dataset contains 458,708 blocks; 208,255 have at least one operable treatment chain and form the search universe. The horizon is 200 years, divided into twenty 10-year accounting periods. The initial schedule is the oldest-first seed from the artifact. It is feasible but is not an optimised control; its objective is 1,722,140,000, dominated by even-flow deviations.

The target combines the weight-10,000 even-flow band with the weight-100 level term of Equation 4. The 70M m^3 per-period aspiration is above the attainable level by design: it is a pull-up gradient, not a feasibility constraint. The reported run has no hard spatial constraints; it is not a test of adjacency or opening-size feasibility. A reader can obtain the same blocks, target, and initial schedule, and the compiled build is a derived artifact with its own terms. The instance is large enough to exercise the compiled boundary and small enough to run a 100,000-generation campaign on one workstation. That combination is why it is used, not a claim that TSA24 represents all timber supply areas.

3.2. Scaling from 1,000 to 100,000 generations

Table 1 reports the seed-7 runs at three budgets. The 1,000-generation run reduced the objective from 1,722,140,000 to 8,588,730 in 9.38 s of search time (23.1 s wall). The 10,000-generation run reached 76,540.7 in 89.8 s search (157 s wall). The 100,000-generation run reached 517.862 in 856 s search (942 s wall). Wall time includes model and cache setup; search time is the GPU budget.

The objective falls by roughly two orders of magnitude from 1,000 to 10,000 generations and by another two orders from 10,000 to 100,000 (Figure 2). That is not a convergence proof; it shows that the search is

Table 1
Seed-7 scaling on the frozen TSA24 instance. Wall time includes model and cache setup; search time is the GPU search budget.

Generations	Objective	Flow term	Level term	Search / wall (s)
1,000	8,588,730	8,588,230	499.821	9.38 / 23.1
10,000	76,540.7	76,026.9	513.832	89.8 / 157
100,000	517.862	2.721	515.141	856 / 942

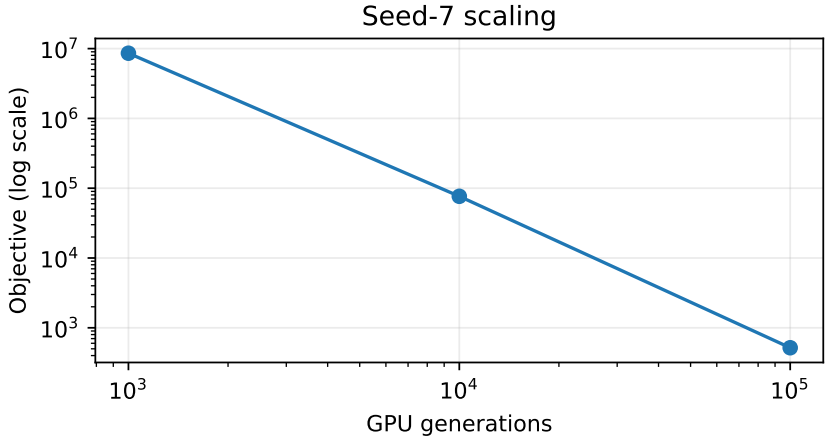


Figure 2: Seed-7 scaling from 1,000 to 100,000 GPU generations. Both axes are logarithmic.

still making progress at the largest budget. The flow term collapses faster than the level term, because the even-flow target is what the population can improve by reshaping the profile. The level term remains between 499.821 and 517.290 because the aspiration is above the achievable level, not because the search ignores it. A 1,000-generation run is a smoke test and a scaling anchor, not a competitive result; the three budgets make that sensitivity visible.

3.3. Five-seed campaign

Table 2 reports the five-seed campaign at 100,000 generations. The best objective is 512.574, the worst is 518.621, the mean is 515.682, and the median is 515.938. Seed 7, the scaling reference, lies at 517.862 and is not part of the five-seed summary. All five campaign schedules replay exactly, and the campaign took 4.00×10^3 s wall for all five runs.

The realized level is approximately 51.21–52.49M m^3 per 10-year period against the 70M aspiration. The level term is therefore the dominant part of the final objective, not a residual that the search failed to remove. The initial seed is dominated by flow deviations; the final schedules have almost no flow penalty but still pay for the level aspiration. Reporting the split prevents a reader from inferring that the target was nearly satisfied. The five-seed spread is small in objective, but the seeds do not all reach the same flow/level compromise. The mean and median are descriptive statistics for five runs, not a confidence interval or a test against a baseline.

3.4. Evaluation path and batch microbenchmark

A full-instance whole-schedule ablation is impractical, because a single full evaluation of the 208,255-block schedule is the motivating cost. We therefore report a disclosed diagnostic on the first 5,000 blocks of the recorded 100,000-generation seed-7 schedule, over 50 moves. A cold whole-schedule evaluation took 54.8 s;

Table 2
Five-seed 100,000-generation campaign. All objectives are exact CPU replays; the mean and median in the text use these five seeds.

Seed	Objective	Flow term	Level term	Realized level (M m^3/period)
1	518.621	1.331	517.290	51.21–52.25
2	512.574	0.000	512.574	51.46–52.49
3	515.938	0.000	515.938	51.31–52.32
4	514.746	0.000	514.746	51.34–52.35
5	516.530	0.076	516.454	51.26–52.29

Table 3
Evaluation-path diagnostic on the first 5,000 blocks of the recorded 100,000-generation seed-7 schedule, 50 moves. Times are exact CPU measurements; relative time is warm whole-schedule move time divided by the row time.

Path	Measurement	Time	Relative	Error
Whole schedule, cold	first evaluation	54.8 s	–	–
Whole schedule, warm	mean per move	1.09 s	1	–
Incremental delta + apply	mean per move	0.542 ms	2.00×10^3	3.7×10^{-9}

after warm-up, a whole-schedule move evaluation took 1.09 s. Exact incremental delta plus apply took 0.542 ms per move, a factor of 2.00×10^3. The final incremental objective matched the whole-schedule objective to 3.7×10^{-9} (Table 3). The subset is not a random sample, and the 50 moves are a diagnostic sequence rather than a search trajectory. The result isolates the evaluation path from the GPU population search; it is not a full-instance ablation and does not measure a matched CPU search.

A second check evaluates the same 64-genome batch against four mixed feature/product cells on CPU and CUDA, using the TSA24 factorized archive. The CPU batch takes 23.1 s and the CUDA batch 0.0931 s, a factor of 248; throughput is 2.77 and 687 genomes/s. The maximum absolute error against the exact evaluator is 1.71×10^{-6} on both paths. The factorized build takes 234 s and the GPU peak allocation is 9.27 GB. This is a batch microbenchmark, not an end-to-end search comparison and not evidence that the architecture causes a search-speed gain. The CPU path is a single-device tensor path, not a tuned multithreaded batch, and the build cost matters only when the table is reused.

3.5. What the validation establishes

The results establish that the software can run a 208,255-operable-block instance at a 100,000-generation GPU budget and produce exact, replayable schedules; that the sparse table plus resident state avoids a full re-simulation without changing the objective; and that the factorized GPU batch evaluates a population quickly on this instance. They do not establish a controlled architecture-versus-implementation effect, an independent quality comparison, a tuned CPU/GPU ranking, or optimality. The retained CPU simulated-annealing run is untuned, single-seed, and not equal-work; it is not used here as a quality baseline.

4. Illustrative example

A user can run the public snapshot directly: extract the TSA24 release tarball, clone WS4, set `WS4_ROOT`, `TSA24_ROOT`, and `FHOPS_ROOT`, then run

```
WS4_ROOT=/path/to/ws4 TSA24_ROOT=/path/to/tsa24 \
FHOPS_ROOT=/path/to/fhops bash paper/replay_all.sh
```

The expected result is `VALID` for every record; the TSA24 records report the stored objective to within 10^{-6}.

5. Impact

The immediate software impact is measured in exact replay and reduced evaluation cost, not in adoption numbers: a 100,000-generation campaign is replayable from stored assignments, and a move evaluation drops from 1.09 s to 0.542 ms on the diagnostic subset. The architecture is reusable when the model boundary is the expensive part of a project. A team with a compiled model and an exact evaluator can add a candidate index and a sparse effect table without changing model semantics. Because weights are data, a sensitivity study can reuse the table and recompute only pricing. Adapting to a new model requires a compiler that emits the tables and manifest, a candidate-index builder for the model's eligibility rules, and an effect-row builder that maps a block/chain pair to target cells. The state, search loop, record schema, and replay gate are designed to be model-agnostic.

5.1. Reuse potential

5.2. Limitations

The limitations are explicit. The validation covers one landscape, one target construction, and one 20-period horizon. The reported run has no hard adjacency or opening-size constraints, so the software is not validated on the spatial-constraint case that motivates much of the literature. There is no independent harvest-scheduling model or LP/MIP bound on the same instance, so the objective values are not a competitive quality result. The five-seed campaign is descriptive; the evaluation-path ablation uses a 5,000-block subset; and the factorized result is a 64-genome batch microbenchmark.

Two constraints follow. The effect table must fit in device memory alongside the population batch; the campaign required a multi-gigabyte device allocation, and a larger candidate set would need a different partitioning. The GPU code is public as release candidate `ws4-v0.2.0-p14rc1`; the paper-specific schedules and replay scripts are in the public papers repository under release candidate `p14-evidence-rc9`. The release is a public snapshot; a DOI is still pending, so the replay contract is not yet an archival DOI service. The target construction is also a policy choice: the software can make it exact and replayable, not correct, and the TSA24 configuration is not an operational decision.

6. Conclusions

WS4 turns a compiled landscape model into a stable sparse representation, maintains exact resident accounting, searches with a batched GPU population, and replays every reported schedule through an exact CPU evaluator. On the public TSA24 instance, the software completes a 100,000-generation, five-seed campaign on 208,255 operable blocks, and all five incumbents replay exactly. The diagnostic measurements show that a move can be priced from a sparse effect row instead of a whole-schedule re-simulation, with objective agreement at 3.7×10^{-9}.

The contribution is the software architecture and its evidence boundary, not a new metaheuristic or a landscape-quality claim. The remaining release steps are a DOI and a tested container replay, a tuned CPU comparison at equal work, a full-instance evaluation-path study, and a spatial-constraint extension. Until those exist, the software is an exact, replayable platform for controlled landscape-search experiments, not a leaderboard entry. It is a reproducible case, not a competitive solver.

CRediT authorship contribution statement

Gregory Paradis: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization, Project administration.

Funding

No specific grant from a funding agency supported this work.

Competing interests

The author declares no competing interests.

Nr.	Code metadata description
C1	Current code version: `ws4-v0.2.0-p14rc1` (public prerelease)
C2	Permanent link to code/repository: https://github.com/UBC-FRESH/ws4/releases/tag/ws4-v0.2.0-p14rc1
C3	Public release capsule (prerelease; DOI pending): https://github.com/UBC-FRESH/fresh-solutions-papers/releases/tag/p14-evidence-rc9
C4	Legal code licence: Apache-2.0 for WS4; the TSA24 artifact retains its own public terms
C5	Code versioning system: Git
C6	Software code languages and tools: Python 3.12; NumPy; PyTorch (CUDA); Numba (optional)
C7	Compilation requirements, operating environments, and dependencies: Linux; Python 3.12.3; CUDA-capable GPU with multi-gigabyte memory (9.27 GB peak in the factorized batch); virtual-environment install
C8	Link to developer documentation: https://github.com/UBC-FRESH/ws4/tree/main/docs
C9	Support email: mailto:gregory.paradis@ubc.ca

Table 4
Code metadata (mandatory).

Code metadata

Data and code availability

The TSA24 benchmark artifact is available at https://github.com/UBC-FRESH/femic-tsa24-instance/releases/tag/ws4-tsa24-evenflow-v1. The build and replay recipe is in `BENCHMARK_ARTIFACT.md` in that repository. The paper-specific schedules and replay scripts are in the public evidence release https://github.com/UBC-FRESH/fresh-solutions-papers/releases/tag/p14-evidence-rc9; a DOI remains pending.

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